Variance Reduction and Importance Sampling: Recap

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The rate is fixed; the constant is not

Every estimator today is still a sample mean, so nothing changes here:

$$\hat{\theta}_n = \frac{1}{n}\sum_{i=1}^{n} h(X_i), \qquad \mathrm{Var}\left(\hat{\theta}_n\right) = \frac{\tau^2}{n}, \qquad \tau^2 = \mathrm{Var}\left(h(X)\right)$$

$$\text{error} = \tau\, n^{-1/2} = O(n^{-1/2})$$

- n buys only the **rate**, and it is expensive: one more digit of precision costs $100\times$ the draws.
- τ is the **constant**, and it is ours to choose. Two estimators of the same θ differ *only* in τ; the smaller one is **more efficient**.

Antithetic variables

Pair each draw with a partner having the **same distribution** and **negative correlation**:

$$\hat{\theta} = \frac{1}{m} \sum_{i=1}^{m} \frac{h(V_i) + h(V_i')}{2}$$

- With monotone h: Uniform: $V' = 1 - V$. Symmetric: $V' = -V$.
- On a matched budget of $2m$ evaluations, with $\rho = \mathrm{Cor}\,(h(V), h(V'))$:

$$\textbf{percent variance reduction} = -\rho$$

- The pair is the unit of independence — you have m independent things, not $2m$.

Importance sampling

Sample from an **envelope** g instead of f, and reweight:

$$\theta = \int h(x)f(x)\,dx = \mathsf{E}\left(h(Y)\frac{f(Y)}{g(Y)}\right), \qquad Y \sim g$$

- $f(Y)/g(Y)$ are the **importance weights**.
- The variance-minimizing envelope is imitates the shape of $|h(y)|f(y)$:

$$g(y) \propto |h(y)|\, f(y)$$

- Two requirements: $g > 0$ wherever $hf \neq 0$, and the weights bounded, $hf/g < c$.
- **The envelope's tails must be at least as heavy as the target's.** Lighter $\Rightarrow$ unbounded weights $\Rightarrow$ infinite variance, and no n fixes it.

It is the tool of choice for **tail probabilities**, where direct sampling puts almost no draws where they matter.